Sampling

1 What Is Aliasing?

Suppose we are analyzing a continuous-time signal containing the frequency component $e^{i\omega_0 t}$. If we sample this signal at a fixed detector every T seconds, we will see the component as a sequence $e^{i\omega_0 Tn}$. Could there exist a frequency component which would generate the same sample sequence if sampled with interval T?

We will assume that such a frequency component does exist, with frequency ω_1. Then

$$e^{i\omega_1 Tn} = e^{i\omega_0 Tn}$$

$$\omega_1 T = \omega_0 T + 2\pi m$$

$$\omega_1 = \omega_0 + \frac{2\pi m}{T}$$

$$f_1 = f_0 + f_s m, \quad m \in \mathbb{Z} \tag{1}$$

Thus, any component with frequency f_1 satisfying (1) will generate the same sample sequence as the original component with frequency f_0.

As demonstrated above, sampling gives an ambiguous view of the frequency content of a continuous-time signal: each sampled sequence could have arisen from any of the infinite frequencies satisfying (1). This loss of information about the true frequency content of a sampled signal is called aliasing because components masquerade as signals at frequencies other than the true frequency which created them.

2 How Can Aliasing Be Avoided?

When sampling, we are interested in knowing the true frequency content of a signal. Without constraints on the spectrum of the signal, we would have to use an infinite sampling frequency as shown in (1) to completely prevent aliasing. This is obviously not practical.

We can prevent aliasing if we constrain the spectrum of the signal being sampled. If the signal contains a limited band of frequencies, then the number of possible aliases for each frequency component of the sampled sequence can be reduced to one by choosing a sufficiently large sampling frequency.

It is most common to limit the spectrum of a signal using a lowpass filter. If we know that the highest frequency component in a signal occurs at cutoff frequency f_c, then, by (1), there will only be one possible alias for the frequency if $f_s > 2f_s$.

This is the essence of the Nyquist-Shannon Sampling Theorem, which states that a band-limited signal with a maximum frequency component f_c can be sampled without aliasing if the sampling frequency f_s is chosen so that $f_s > 2f_c$.